

# Dual ordinal spectral ranking of journals and institutions

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April 6, 2026

## Abstract

We develop a dual spectral ranking of journal and institution units that determines influence-per-work scores via the block citation matrix. A binary block-mask selects among four bibliometrically distinct flow structures, nesting single-unit rankings (such as the eigenvector journal ranking) alongside two forms of journal-institution coupling. Applied to the economics and commerce journal literature using OpenAlex data, the method produces stable, interpretable joint rankings whose sensitivity to model parameters is examined systematically.

**Keywords:** spectral ranking, dual journal-institution prestige, Katz-Hubbell resolvent

## 1 Introduction

The research system relies on journals and institutions. Journals filter works through editorial and peer review, archive the literature, and create metadata for retrieval and exploration. Institutions fund individual researchers, implement training, oversee compliance, establish research organisational units and maintain physical and information infrastructure. Although the status hierarchy that links journals and institutions is a pervasive aspect of the research system (Merton, 1973; McNamee and Willis, 1994; Martin, 1998), bibliometric studies that explore the joint hierarchy of journals and institutions are rare. This paper proposes a joint ordinal spectral ranking of journals and institutions derived from bibliometric data. The method helps to reveal how status is formed and maintained. We also intend to apply the approach to study the relations between the status of journals and institutions, and researcher esteem.

The introduction of the science citation index (Garfield, 1964) opened a new way to investigate connections between journals and institutions. Citation-based journal ranking in particular became a useful means to manage collections and to select outlets for publications. On the other hand, the misapplication of the journal impact factor (JIF) as proxy for the importance of research works published in that journal fostered a still-unresolved tension in bibliometric evaluation (Garfield, 1979; MacRoberts and MacRoberts, 2018). Perhaps due to the complexity of disciplines and missions, an institution impact factor (IIF) has not emerged. Rather, institutional rankings derived from bibliometric indicators of works, authors or journals are often composed with other indicators to form an institutional score. This is the approach in global rankings such as QS, Times Higher Education (THE) and Shanghai Ranking (ARWU), the academic Leiden and SCImago Institutions rankings, as well as many national performance-based research funding schemes (Zacharewicz et al, 2019).

The prospects for using a citation index to explore research importance were considered in an early work by Narin (1976), Pinski and Narin (1976), and Narin et al (1976) who proposed a recursive method to construct influence weights of journals within a disciplinary field. To obtain influence weights for institutions, Narin added publications weighted by the respective journal influence score. Over a sample of several dozen universities, the total institution influence scores were well-correlated with survey-based rankings, but also with the number of publications, suggesting that survey respondents combined influence and size. Narin's work built on earlier studies of social networks and developed into the eigenvector approach now widely used for journal ranking (Bergstrom et al, 2008; Vigna, 2016; Franceschet, 2010). Eigenvector methods have also been applied to author-level influence weightings in the field of social science (West et al, 2013), and extended by Narin's approach to ranking institutions and countries. An alternative approach iterates between the article-article network linked by the reference lists (without projection) and the corresponding article-author network (Ujum et al, 2015; Pal and Ruj, 2015). The formulation by Cao et al (2023) adopts a similar network, with reference-based article-article links, and the related article-institution links, ranked by a novel and effective iteration over random walks. This is one of very few spectral rankings of institutions. The authors note that a limiting case of this approach corresponds to the eigenvector (pageRank) method.

As noted earlier by Geller (1978), random walks over Markov chains are closely related to the spectral ranking problems discussed in this paper.

It is useful to foreground analytical devices adopted in this paper. First, we say that a *reference* is an entry  $w_2$  in the reference list of a work  $w_1$ :  $w_1$  references  $w_2$  and  $w_2$  is referenced by  $w_1$ . Bibliometric indexers establish the connection between  $w_2$  and the entry for  $w_2$  in an index of the *citations* to  $w_2$  (Wouters, 1999) we say  $w_1$  cites  $w_2$  and  $w_2$  is cited by  $w_1$ . A work can be fractionated among its authors and institutions, signifying the fraction of  $w_1$ 's *attention* given (paid) to the authors and institutions of  $w_2$ . Spectral ranking accounts for the weakening and strengthening of  $w_1$ 's attention by the attention paid to it by others, then distributed by fractionation. We can also say that a citation signifies the *influence* of  $w_2$ 's authors and institutions on  $w_1$ ; like attention, influence can be weakened or strengthened by other citations to  $w_2$ . Second, in this paper we view references and citations through Social Systems Citation Theory (Tahamtan and Bornmann, 2022, SSCT). We concentrate on the structural features of the network of references and citations, setting aside authorial intent. This approach contrasts with actor-network theory (Latour, 2007, ANT), which foregrounds the intentful roles of heterogeneous agents in assembling citation networks. ANT will be important in our follow-up studies that relate this paper to the referencing activities of individual researchers.

This paper proceeds as follows. In the next section, we construct a framework for computing the transfer of attention or influence among journal and institution units coupled by the reference lists of the works they publish. We then construct a corpus of works in the fields of Economics and Business, forming subset of the corpus to illustrate the interconnections of units, fields, and time. We present spectral rankings over these subsets, alongside sensitivity and bootstrap studies of some key parameters. We then discuss the results, including a critique of our methods, and close by mentioning some directions for future work.

## 2 Journal–institution citation networks

We begin by selecting  $N_s$  sources  $\mathfrak{S}$  in a field, discipline or category. Then we select  $N_w$  works  $\mathfrak{W}$  published in  $\mathfrak{S}$  within the census window years  $t_c = [t_0^c \dots t_1^c]$  or target years  $t_t = [t_0^t \dots t_1^t]$ . From the reference lists of the set of census works  $\mathfrak{W}^c$  we identify the set of target works  $\mathfrak{W}^t$  and form the filtered reference set  $\mathfrak{R} = \{(i, j) : w_i \in \mathfrak{W}^c, w_j \in \mathfrak{W}^t\}$ . Author affiliations in  $\mathfrak{W}$  establish a set of authors  $\mathfrak{A}'$  and institutions  $\mathfrak{U}'$ . In practice many institutions (and sources) publishing only a few works per year can be dropped from the corpus. Let  $\bar{w}_u$  be the mean annual census-window work count for institution  $u$  and  $\bar{w}_s$  the corresponding count for source  $s$ . We determine lower limits  $\tau_U$  and  $\tau_S$  by examining the respective frequency distributions and define  $\mathfrak{U} \subseteq \mathfrak{U}'$  as the institutions with  $\bar{w}_u \geq \tau_U$ , and  $\mathfrak{S} \subseteq \mathfrak{S}'$  as the sources with  $\bar{w}_s \geq \tau_S$ , with corresponding author set  $\mathfrak{A} \subseteq \mathfrak{A}'$ . Works that lose all their authorships under the institution filter, or whose source is dropped by the source filter, are removed from the corpus. Each retained work  $w_i$  has an *institution weight* under author-fractional counting (Waltman and van Eck, 2015),

$$\omega_{iu} = \sum_{\ell: u \in \cap_{i\ell}} \frac{1}{a_i} \frac{1}{u_{i\ell}}, \quad (1)$$

where  $\ell$  indexes the retained authors of work  $w_i$ ,  $a_i$  is their count, and  $u_{i\ell}$  the number of retained institutions of author  $\ell$ , so that  $\sum_u \omega_{iu} = 1$ . Re-weighting fractional counts as a consequence of pruning the number of institutions is a trade-off between boosting a work's remaining institutions and diminishing the work's total contribution.

Each reference  $(i, j) \in \mathfrak{R}$  is a transaction between works: the referencing work gives, and the referenced work receives. Correspondingly, the units (source and institutions) of the referencing work transact with the units of the referenced work, according to weights defined in the citation matrix  $\mathbf{C}$  defined below.

We attach the construct *attention* to these unit-to-unit transactions. Initially the amount transacted is the fractionated share given by sources and the the  $\rho_i$ -weighted fractional authorship accumulated in  $W$  then assert that the attention a unit gives should itself be amplified or diminished by the attention that unit has received: a reference from a highly-attended unit carries more weight than one from a lightly-attended unit. This leads directly to the eigenvector equation, in which the attention of each unit is defined recursively as the weighted sum of the attention of the units that reference it, with its value established only by the convergence of that recursion. The process of finding the fixed point is *spectral ranking* (a verb); the resulting

vector is the *spectral ranking* (a noun). The vector can be read from either side of the transaction — from the referencing side as attention, or from the cited side as influence — just as  $\mathbf{C}$  can be called a reference matrix or a citation matrix depending on which side one stands.

A work with many references distributes its attention thinly across its reference list; one with few concentrates it. Let  $R_i = |\{j : (i, j) \in \mathfrak{R}\}|$  be the reference count of work  $w_i$  and  $\bar{R} = |\mathfrak{R}|/N_w$  the mean reference count over  $\mathfrak{W}$ . The *attention weight*  $\rho_i$  controls the total attention each work contributes, balancing the amounts of attention across works (Zitt and Small, 2008):

$$\rho_i = \begin{cases} 1 & \text{(full count: attention proportional to reference count)} \\ \bar{R}/R_i & \text{(fixed count: equal total attention per work).} \end{cases} \quad (2)$$

Under full count a work with twice as many references contributes twice as much total attention; under fixed count every work contributes the same total attention  $\bar{R}$ , so that prolific citers do not dominate the matrix.

We construct the set of *units* [sources or institutions; Pinski and Narin (1976)]  $\mathfrak{P} = \mathfrak{S} \cup \mathfrak{U}$  with  $N = N_s + N_u$  elements, ordered so that sources occupy indices  $1, \dots, N_s$  and institutions indices  $N_s + 1, \dots, N$ . Each reference  $(i, j) \in \mathfrak{R}$  carries attention from the units associated with  $w_i$  to the units associated with  $w_j$ , across four unit-type pairs: source–source (*SS*), source–institution (*SI*), institution–source (*IS*), and institution–institution (*II*).

For each work  $w_i$  we define the source membership one-hot vector  $\mathbf{b}_i^S \in \mathbb{R}^{N_s}$  with  $[\mathbf{b}_i^S]_s = \mathbf{1}[w_i \in s]$ , and the institution weight distribution vector  $\mathbf{b}_i^U \in \mathbb{R}^{N_u}$  with  $[\mathbf{b}_i^U]_u = \omega_{iu}$  from (1). The attention carried by each reference is distributed across unit-type pairs and accumulated as four block matrices over  $\mathfrak{R}$  via rank-1 outer products,

$$\mathbf{C}_{XY} = \sum_{(i,j) \in \mathfrak{R}} \rho_i \mathbf{b}_i^X (\mathbf{b}_j^Y)^\top, \quad XY \in \{SS, SI, IS, II\}, \quad (3)$$

where  $\mathbf{b}_i^X$  distributes attention from the citing side across units of type  $X$ , and  $\mathbf{b}_j^Y$  distributes it to the cited side across units of type  $Y$ . Each entry  $[\mathbf{C}_{XY}]_{pq}$  is therefore the total  $\rho_i$ -weighted attention flowing from unit  $p$  of type  $X$  to unit  $q$  of type  $Y$  across all references in  $\mathfrak{R}$ .

The citation matrix assembles the four blocks:

$$\mathbf{C}(\chi, \mathbf{m}, \rho) = \begin{bmatrix} m_{SS}(1-\chi)^2 \mathbf{C}_{SS} & m_{SI}\chi(1-\chi) \mathbf{C}_{SI} \\ m_{IS}\chi(1-\chi) \mathbf{C}_{IS} & m_{II}\chi^2 \mathbf{C}_{II} \end{bmatrix}, \quad (4)$$

where the binary *block mask*  $\mathbf{m} = (m_{SS}, m_{SI}, m_{IS}, m_{II}) \in \{0, 1\}^4$  is bibliometrically meaningful for source-only (1000), institution-only (0001), bipartite (0110), and full-joint (1111) combined blocks. The mixing weight  $\chi$  controls the source–institution balance in the bipartite and full-joint cases and is inert in the single-mode cases. The  $\chi$  scaling preserves total attention flow across all block combinations, since  $(1-\chi)^2 + 2\chi(1-\chi) + \chi^2 = 1$ .

Note that the corpus excludes (a) census work references not giving attention to target works, (b) non-census works giving attention to target works, and (c) authorships that include an omitted institution.

**Ordinal spectral ranking.** Let  $r_p = \sum_q C_{pq}$  be the total weighted attention given by unit  $p$  across all its references, and  $\mathbf{D}_r = \text{diag}(r_p)$  the corresponding diagonal matrix. Row-normalising  $\mathbf{C}$  to obtain  $\mathbf{H} = \mathbf{D}_r^{-1} \mathbf{C}$  ensures that each unit’s references are equally valued regardless of how many references it makes: a unit with many references exerts no more attention per reference than one with few. Size dependence is deliberately removed at this stage and reintroduced only at the end via the influence per work scaling (7). If  $r_p = 0$  (a dangling unit with no outgoing references) the corresponding row of  $\mathbf{H}$  is defined to be zero.

The three unipartite masks—source-only (1000), institution-only (0001), and full-joint (1111)—each yield a square  $\mathbf{H}$  to which the following spectral ranking theory applies directly. The spectral ranking  $\boldsymbol{\pi}^*$  (see Vigna, 2016, for an overview) is the unique positive solution to the eigenvector equation,

$$(\mathbf{I} - \mathbf{H}^\top) \boldsymbol{\pi}^* = \mathbf{0}, \quad \mathbf{1}^\top \boldsymbol{\pi}^* = 1, \quad \boldsymbol{\pi}^* \geq 0, \quad (5)$$

when  $\mathbf{H}$  is primitive (irreducible and aperiodic), as guaranteed by the Perron–Frobenius theorem. Our implementation tests primitivity for every  $\mathbf{H}$  before applying the power iteration and has not encountered

a failure. Geller (1978) shows that  $\pi^*$  is the stationary distribution of the Markov chain with transition matrix  $\mathbf{H}$ , and that it coincides with the *influence weight* of Pinski and Narin (1976) scaled by the reference count proportion  $r_p / \sum_q r_q$ .

Katz (1953) proposed attenuating influence along walks of length  $k$  by a factor  $\alpha^k$ ,  $\alpha \in (0, 1)$ , so that distant endorsements count for less than proximate ones. Summing attenuated walks of all lengths and endowing every unit with a uniform initial score of 1 yields the resolvent equation

$$(\mathbf{I} - \alpha \mathbf{H}^\top) \pi = \mathbf{1}.$$

Since  $\mathbf{H}$  is row-stochastic,  $(\mathbf{I} - \alpha \mathbf{H}^\top)$  is invertible for any  $\alpha \in (0, 1)$ , guaranteeing a unique solution.

Hubbell (1965) generalised this by replacing the uniform endowment  $\mathbf{1}$  with an informative prior  $\mu$  over units, representing exogenous information about standing independent of citation structure,

$$(\mathbf{I} - \alpha \mathbf{H}^\top) \pi = \mu.$$

We use the uniform prior  $\mu_p = 1/N_{\mathfrak{P}}$  for  $p \in \mathfrak{P}$ . The case  $(\mathbf{I} - \alpha \mathbf{H}^\top) \pi = \mathbf{0}$  with  $\alpha \neq 1$  admits only the trivial solution  $\pi = \mathbf{0}$  and is not useful as a ranking.

Combining path attenuation and prior information yields the general resolvent equation

$$(\mathbf{I} - \alpha \mathbf{H}^\top) \pi = (1 - \alpha) \mu, \quad \mathbf{1}^\top \pi = 1, \quad \pi \geq 0, \quad (6)$$

which recovers  $\pi^*$  in the limit  $\alpha \rightarrow 1$  when  $\mathbf{H}$  is primitive. While  $\alpha$  and  $\mu$  are introduced here for their substantive interpretations—path attenuation and exogenous prior respectively—they have the additional effect of ensuring a unique solution to (6) even when  $\mathbf{H}$  is not primitive:  $\alpha < 1$  makes  $(\mathbf{I} - \alpha \mathbf{H}^\top)$  invertible, and a nonzero  $\mu$  lifts the equation off the homogeneous system.

The spectral ranking  $\pi$  reflects the total influence of each unit and depends on unit size. The *influence per work* (Palacios-Huerta and Volij, 2004),

$$\mathbf{v} = A \mathbf{A}^{-1} \pi, \quad \mathbf{A} = \text{diag}(a_p), \quad A = \mathbf{a}^\top \mathbf{1} = \sum_p a_p, \quad (7)$$

removes this dependence, where the unit work count  $a_p = |\{i : w_i \in p\}|$  where  $p$  is a specific source or institution, and  $\pi$  denotes the solution to (6). The scalar  $A$  normalises  $\mathbf{v}$  so that its  $a_p$ -weighted mean equals one:  $\mathbf{a}^\top \mathbf{v} / A = \mathbf{1}^\top \pi = 1$ . A unit with  $v_p = 1$  therefore has the corpus-average influence per work and  $v_p = 2$  indicates twice the average influence per work. Palacios-Huerta and Volij (2004) show that  $\mathbf{v}$  is uniquely characterised by several reasonable axiomatic constraints (but see Serrano, 2004).

**Bipartite ranking.** With  $\mathbf{m} = (0110)$ , equation (4) yields block off-diagonal matrices  $\mathbf{C}$  and  $\mathbf{H}$ , forming a directed bipartite network between sources ( $S$ ) and institutions ( $I$ ). This is the mask used as the baseline model in Section 4. Partitioning rows and columns into the  $S$  and  $I$  blocks, the stochastic matrix takes the form

$$\mathbf{H} = \begin{pmatrix} \mathbf{0} & \mathbf{H}_{SI} \\ \mathbf{H}_{IS} & \mathbf{0} \end{pmatrix}, \quad (8)$$

where  $\mathbf{H}_{SI}$  distributes a source's attention across institutions and  $\mathbf{H}_{IS}$  does the reverse. Because every path alternates  $S \rightarrow I \rightarrow S \rightarrow \dots$ , this structure makes  $\mathbf{H}$  periodic with period 2; the resolvent equation (6) ensures a unique solution for any  $\alpha \in (0, 1)$ .

Squaring  $\mathbf{H}$  collapses the two-step walks back onto each node type, giving the block-diagonal one-mode projections

$$\mathbf{H}^2 = \begin{pmatrix} \mathbf{M}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_I \end{pmatrix}, \quad \mathbf{M}_S = \mathbf{H}_{SI} \mathbf{H}_{IS}, \quad \mathbf{M}_I = \mathbf{H}_{IS} \mathbf{H}_{SI}.$$

Both  $\mathbf{M}_S$  and  $\mathbf{M}_I$  are row-stochastic products of row-stochastic matrices.

To make the bipartite mode directly comparable with the unipartite SS and II modes at the same  $\alpha$ , each single arc  $SI$  or  $IS$  is damped by  $\sqrt{\alpha}$ , so every round-trip through  $\mathbf{M}_S$  incurs attenuation  $\sqrt{\alpha} \cdot \sqrt{\alpha} = \alpha$ , matching the single-step damping elsewhere. Substituting  $\sqrt{\alpha}$  for  $\alpha$  in equation (5), partitioning  $\pi = (\pi_S, \pi_I)$ , and eliminating  $\pi_I$  yields the bipartite resolvent for sources,

$$(\mathbf{I} - \alpha \mathbf{M}_S^\top) \pi_S = (1 - \sqrt{\alpha}) (\mu_S + \sqrt{\alpha} \mathbf{H}_{IS}^\top \mu_I), \quad (9)$$

where the right-hand side transports institution prior mass back to sources through  $\mathbf{H}_{IS}^\top$ . The institution scores are then recovered by a single matrix–vector product,

$$\boldsymbol{\pi}_I = \sqrt{\alpha} \mathbf{H}_{SI}^\top \boldsymbol{\pi}_S + (1 - \sqrt{\alpha}) \boldsymbol{\mu}_I. \quad (10)$$

**Computation.** To compute  $\boldsymbol{\pi}$  we define the modified matrix

$$\tilde{\mathbf{H}} = \alpha \mathbf{H} + (1 - \alpha) \boldsymbol{\mu} \mathbf{1}^\top, \quad (11)$$

so that the fixed-point condition  $\boldsymbol{\pi} = \tilde{\mathbf{H}}^\top \boldsymbol{\pi}$  is equivalent to equation (6). Because  $\boldsymbol{\mu} > 0$  and  $0 < \alpha < 1$ , every entry of  $\tilde{\mathbf{H}}$  is strictly positive, ensuring it is a primitive row-stochastic matrix. By the Perron–Frobenius theorem,  $\tilde{\mathbf{H}}^\top$  therefore has a unique stationary distribution  $\boldsymbol{\pi}$ , recovered by power iteration

$$\boldsymbol{\pi}^{(k+1)} = \tilde{\mathbf{H}}^\top \boldsymbol{\pi}^{(k)}, \quad (12)$$

converging to  $\boldsymbol{\pi}$  from any strictly positive starting vector  $\boldsymbol{\pi}^{(0)}$ . The scaling  $A \mathbf{A}^{-1}$  to obtain  $\mathbf{v}$  is applied after  $\boldsymbol{\pi}$  has converged.

$\mathbf{C}(\chi, \mathbf{m}, \rho)$  is a sparse matrix built from a weighted edge list  $\mathcal{E}$  accumulated over  $\mathfrak{R}$  with per-reference weight  $\rho_i$ , stored in CSR format, and row-normalised to  $\mathbf{H}$ . The pipeline  $\mathfrak{R} \rightarrow \mathcal{E} \rightarrow \mathbf{C} \rightarrow \mathbf{H} \rightarrow \boldsymbol{\pi} \rightarrow \mathbf{v}$  is implemented using `scipy.sparse.csr_matrix` with parameter entry points detailed in the Supplement.

### 3 Corpus selection

OpenAlex (Priem et al, 2022) provides our bibliometric data. Corpus selection starts with a list of eligible sources having four features: an OpenAlex source identifier for corpus extraction, evidence from one of our journal registries that the source publishes economics or business research, a sufficient number of published works per year, and relatively high OpenAlex source topic density to exclude cross-disciplinary journals.

The journal registries are as follows. Harzing’s *71st Edition of the Journal Quality List* (Harzing, 2024, JQL) assembles peer assessments of journal quality across business and economics disciplines. Its scope is confined to economics and business so no filtering is applied. The *ERA 2023 Submission Journal List* (Australian Research Council, 2022, SJL) assigns up to three Field of Research (FoR) codes to each entry. From the full list of 26,241 journals we retain those having all FoR codes falling within Division 35 (Commerce, Management, Tourism and Services) or Division 38 (Economics). This applies to two-digit ‘division’ codes and all four-digit ‘group’ codes (3501–3509, 3801–3803, 3899). After filtering, 1,602 journals remain. *Clarivate Web of Science Journal Citation Reports* (Clarivate Analytics, 2026, JCR) assign each journal a Web of Science category. We retain those in Economics, Management, Business, Transportation or Business, Finance. After filtering, 1,413 journals remain.

Every journal in JQL, SJL, and JCR has at least one ISSN, and may have several. These registry ISSNs are joined independently to the February 2026 OpenAlex sources snapshot using its list of ISSNs as the matching key and selecting only the ‘journal’ source-type. Journal titles are examined to identify around 30 additional matches where the OpenAlex source does not have an ISSN. The list of OpenAlex sources that are matched is designated OAS\*.

The long list OAS\* includes sources publishing many works in disciplines beyond economics and business, and we use OpenAlex source-topic assignments to exclude them as follows. A source has work-topic counts for up to five topics, each within a single sub-field, field, and domain. We accumulate the work-topic counts for OpenAlex Field 14 (Economics, Econometrics and Finance) and Field 20 (Business and Management) and drop sources if this count is below 50% of the total count.

For the surviving OAS\* sources we extract work and authorship data revealing units that can be excluded because they publish very few works. To find a suitable threshold for excluding these units, we examine source and institution retention curves (Fig. 1), selecting thresholds of  $\tau_U = 20$  works/institution/year and  $\tau_S = 20$  works/source/year. We designate the residual set of sources OAS.

Using units that satisfy these thresholds, we extract a full corpus of works, authorships and references retaining article and review types with publication years from 2000 to 2024, excluding works that are paratext or retracted, or without an authorship entity, and compute  $\omega_{iu}$  for each work. The characteristics of the full corpus are exhibited in Table 2.

Table 1: Matching of the three register-based sources to OpenAlex source identifiers. Overlaps count shared matched entries; OAS\* is the union across all three registries.

| Register                                     | Journals | Matched | Match rate | Unmatched     |
|----------------------------------------------|----------|---------|------------|---------------|
| JQL                                          | 842      | 835     | 99.2%      | 7             |
| MJL                                          | 1,413    | 1,392   | 98.5%      | 21            |
| SJL                                          | 1,602    | 1,371   | 85.6%      | 231           |
| Overlap: MJL $\cap$ SJL                      |          |         |            | 783 sources   |
| Overlap: MJL $\cap$ JQL                      |          |         |            | 566 sources   |
| Overlap: SJL $\cap$ JQL                      |          |         |            | 422 sources   |
| Overlap: MJL $\cap$ SJL $\cap$ JQL           |          |         |            | 393 sources   |
| (JQL $\cup$ MJL $\cup$ SJL) $\cap$ OA (OAS*) |          |         |            | 2,359 sources |

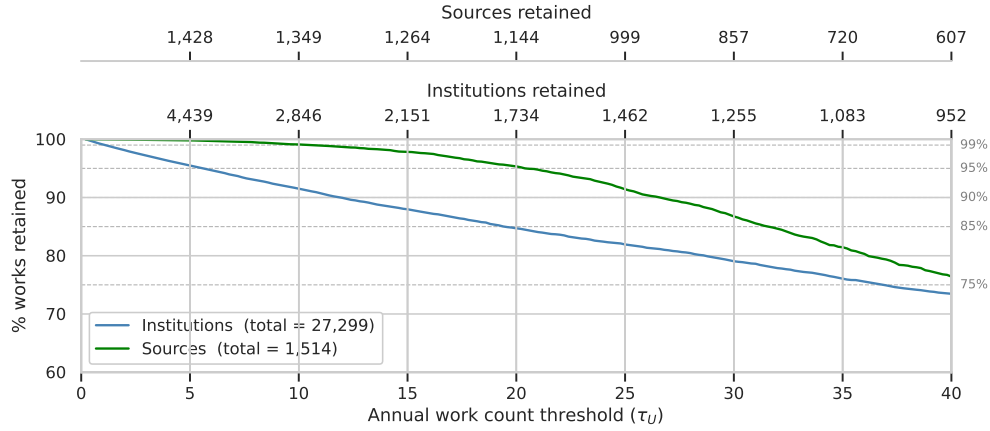


Figure 1: Institution and source retention curves

Table 2: Corpus features by year ( $\tau_U > 20$ ). Upper panel: unique-item counts. Lower panel:  $D_1$  and  $D_9$  deciles of per-entity distributions. The 2000–24 column pools the full period.

| Quantity                         | 2000    |       | 2024      |       | 2000–24    |        |
|----------------------------------|---------|-------|-----------|-------|------------|--------|
|                                  | count   |       | count     |       | count      |        |
| Works                            | 15,011  |       | 51,902    |       | 860,101    |        |
| Sources                          | 664     |       | 1,307     |       | 1,610      |        |
| Institutions                     | 875     |       | 1,010     |       | 1,012      |        |
| Reference counts (out-degree)    | 5,997   |       | 1,378,852 |       | 13,599,071 |        |
| Citation counts (in-degree)      | 560,044 |       | 28,622    |       | 14,044,614 |        |
| Quantity                         | $D_1$   | $D_9$ | $D_1$     | $D_9$ | $D_1$      | $D_9$  |
| Works per source                 | 3.0     | 46.0  | 4.0       | 78.0  | 26.9       | 1191.2 |
| Institutions per work            | 1.0     | 3.0   | 1.0       | 5.0   | 1.0        | 4.0    |
| References per work (out-degree) | 1.0     | 3.0   | 5.0       | 60.0  | 2.0        | 44.0   |
| Citations per work (in-degree)   | 2.0     | 95.0  | 1.0       | 4.0   | 1.0        | 47.0   |

## 4 Results

**Outline.** Tables 3–5 define and enumerate the parameters; all are consolidated here and not repeated elsewhere. The *baseline* is  $t_x = 5$ ,  $F = A$ ,  $\tau_U = \tau_S = 20$ ,  $\rho = 0$ ,  $\mathbf{m} = 0110$ ,  $\alpha = 1$ ,  $\mu = 0$  — the pure eigenvector equation (5) on the bipartite network. Modes  $\mathbf{m} = 1000$  (SS) and  $0001$  (II) are the unipartite analogues of standard journal and institution eigenvector methods; 1111 adds all paths between the two unit types; 0110 is the bipartite focus of this paper. Phase 1 fixes  $\alpha = 1$ ,  $\mu = 0$  and varies the corpus ( $F$ ) and mode ( $\mathbf{m}$ ); Phase 2 varies  $\alpha$ ,  $\rho$ , and  $\tau$ ; Phase 4 varies  $t_x \in \{1, 2, 3, 4, 5\}$ .

Table 3: Model parameters  $\theta$ .

| Symbol                      | Name                    | Type            | Role                                                                                            |
|-----------------------------|-------------------------|-----------------|-------------------------------------------------------------------------------------------------|
| <i>Corpus construction</i>  |                         |                 |                                                                                                 |
| $F \in \{E, B, A\}$         | field subset            | category        | restrict $\mathfrak{S}$ to Economics ( $E$ ), Business ( $B$ ), or all OAS ( $A$ ); see Table 4 |
| $t_x \in \{1, \dots, 5\}$   | time window             | integer         | census and target year ranges; see Table 4                                                      |
| $\tau_U$                    | institution threshold   | integer         | minimum mean annual works to retain an institution                                              |
| $\tau_S$                    | source threshold        | integer         | minimum mean annual works to retain a source                                                    |
| <i>Matrix construction</i>  |                         |                 |                                                                                                 |
| $\rho \in \{0, 1\}$         | reference normalisation | binary          | 0: a work contributes $\bar{R}$ units<br>1: a work contributes $R_i$ units                      |
| $\mathbf{m} \in \{0, 1\}^4$ | block mask              | binary 4-vector | select active unit-type pairs ( $SS$ , $SI$ , $IS$ , $II$ )                                     |
| $\chi \in [0, 1]$           | mixing weight           | continuous      | source–institution balance within each work                                                     |
| <i>Ranking</i>              |                         |                 |                                                                                                 |
| $\alpha \in [0, 1)$         | damping factor          | continuous      | controls propagation through citation network                                                   |
| $\mu$                       | prior weight            | continuous      | exogenous prestige prior; $\mu = 0$ = pure spectral                                             |

Table 4: Corpus construction parameters: field subsets ( $F$ ) and time windows ( $t_x$ ).

*Panel A: Field subsets ( $F$ )*

$F$  Sources included in  $\mathfrak{S}$

$E$  Topic counts exclusively in Field 14 (Economics, Econometrics and Finance)

$B$  Topic counts exclusively in Field 20 (Business, Management and Accounting)

$A$  All OAS sources; topic counts in Fields 14 and/or 20 (*baseline*)

*Panel B: Time windows ( $t_x$ )*

| $t_x$ | Window $[t_0^c, t_1^c] = [t_0^t, t_1^t]$ | Label                                |
|-------|------------------------------------------|--------------------------------------|
| 1     | $t_{00} = [2000, 2004]$                  | Symmetric 5-year                     |
| 2     | $t_{05} = [2005, 2009]$                  | Symmetric 5-year                     |
| 3     | $t_{10} = [2010, 2014]$                  | Symmetric 5-year                     |
| 4     | $t_{15} = [2015, 2019]$                  | Symmetric 5-year                     |
| 5     | $t_{20} = [2020, 2024]$                  | Symmetric 5-year ( <i>baseline</i> ) |

**Phase 1: Sources, institutions and fields.** Figure 2 shows the prestige-per-work distribution for the baseline and for the field subsets  $F = E$  and  $F = B$ , with the  $x$ -axis locked to the baseline rank so that each

Table 5: Parameters and values explored. Baseline values in italics.

| Symbol                     | Name                    | Baseline         | Values explored                  |
|----------------------------|-------------------------|------------------|----------------------------------|
| <i>Corpus construction</i> |                         |                  |                                  |
| $t_x$                      | Time window             | <i>5</i>         | 1, 2, 3, 4, 5                    |
| $F$                        | Field subset            | <i>A</i>         | <i>E</i> , <i>B</i> , <i>A</i>   |
| $\tau_U$                   | Institution threshold   | <i>20</i>        | <i>20</i> , 40                   |
| $\tau_S$                   | Source threshold        | <i>20</i>        | <i>20</i> , 40                   |
| <i>Matrix construction</i> |                         |                  |                                  |
| $\rho$                     | Reference normalisation | <i>0 (fixed)</i> | 0 (fixed), 1 (full)              |
| $\mathbf{m}$               | Block mask              | <i>0110</i>      | 1000, 0001, <i>0110</i> , 1111   |
| $\chi$                     | Mixing weight           | 0.50             | 0.50, $\chi^*$ [full joint only] |
| <i>Ranking</i>             |                         |                  |                                  |
| $\alpha$                   | Damping factor          | <i>1.0</i>       | 0.1, 0.5, <i>1.0</i>             |
| $\mu$                      | Prior weight            | <i>0</i>         | 0 (pure spectral)                |

unit sits at its baseline position and absent units are omitted from that series. Sources at the Economics–Business boundary lose prestige flow from one side under each restriction; the institution panel reflects the same effect through the authorship mapping.

The field shifts in Figure 2 reflect community structure in the citation network: sources cluster into Economics and Business communities that preferentially cite within their own field, and this influences rankings because prestige propagates more readily within a community than across it.

The second eigenpair  $(\lambda_2, \phi_2)$  of  $\mathbf{H}^\top$  characterises this structure precisely. For general  $\alpha$  and uniform prior  $\mu_p = 1/N$ , the solution to (6) has the spectral expansion

$$\boldsymbol{\pi} = (1 - \alpha) \sum_{i \geq 1} \frac{\phi_i^\top \boldsymbol{\mu}}{1 - \alpha \lambda_i} u_i, \quad (13)$$

where  $(\lambda_i, u_i, \phi_i)$  are the eigenvalues, right eigenvectors, and left eigenvectors of  $\mathbf{H}^\top$ , ordered  $\lambda_1 \geq \lambda_2 \geq \dots$ . The  $i = 1$  term recovers  $\boldsymbol{\pi}^*$  as  $\alpha \rightarrow 1$ ; the  $i = 2$  term is the leading community correction, amplified by  $1/(1 - \alpha \lambda_2)$  relative to all higher terms. The spectral gap  $g = 1 - \lambda_2$  therefore governs community influence: small  $g$  means community membership strongly shapes rankings; large  $g$  means rankings are community-neutral. For the SS mode  $\phi_2$  is expected to separate Economics from Business sources; for II the partition is more diffuse because institutions co-locate faculty from both fields. In the bipartite mode the community operator is the one-mode projection  $\mathbf{M}_S = \mathbf{H}_{SI} \mathbf{H}_{IS}$ ;  $\phi_2^S$  is its second eigenvector and the institution community vector is recovered as  $\phi_2^I = \mathbf{H}_{SI}^\top \phi_2^S$  (unit-normalised). Throughout,  $\phi_2$  is signed so that the unit with the largest work count  $a_p$  has  $\phi_{2,p} > 0$ .

Figure 3 illustrates these eigenvectors by plotting the size-adjusted community measure  $\phi_{2,p}/\sqrt{v_p}$  against  $v_p$ ; dividing by  $\sqrt{v_p}$  removes the normalisation artefact that suppresses  $|\phi_{2,p}|$  for low-prestige units.

In the SS source panel (top left), the sign of  $\phi_{2,p}/\sqrt{v_p}$  [FILL: cleanly separates Economics (positive) from Business (negative) journals across the full prestige range, with Finance journals concentrated near zero, confirming their role as citation bridges between the two communities]. The bipartite source panel (top right) shows [FILL: a similar partition / greater mixing], with spectral gaps  $g^{SS} = [\text{FILL}]$  and  $g^{\text{bip}} = [\text{FILL}]$  indicating that routing citation flow through institutions [FILL: partially dissolves / preserves] the community boundary. The institution panels (bottom row) show a more diffuse partition, consistent with institutions co-locating faculty from both fields; the field-network colouring confirms that E-only institutions [FILL: concentrate on one side / are scattered].

Figure 4 compares prestige-per-work distributions across the four network modes, with the  $x$ -axis locked to the bipartite baseline. For sources, modes 1000 (red  $\times$ ) and 1111 at  $\chi^*$  (green  $\times$ ) are overlaid on the bipartite baseline; open circles and squares on the 1000 series mark Economics and Business sources respectively. For institutions, mode 0001 and 1111 replace the source alternatives.



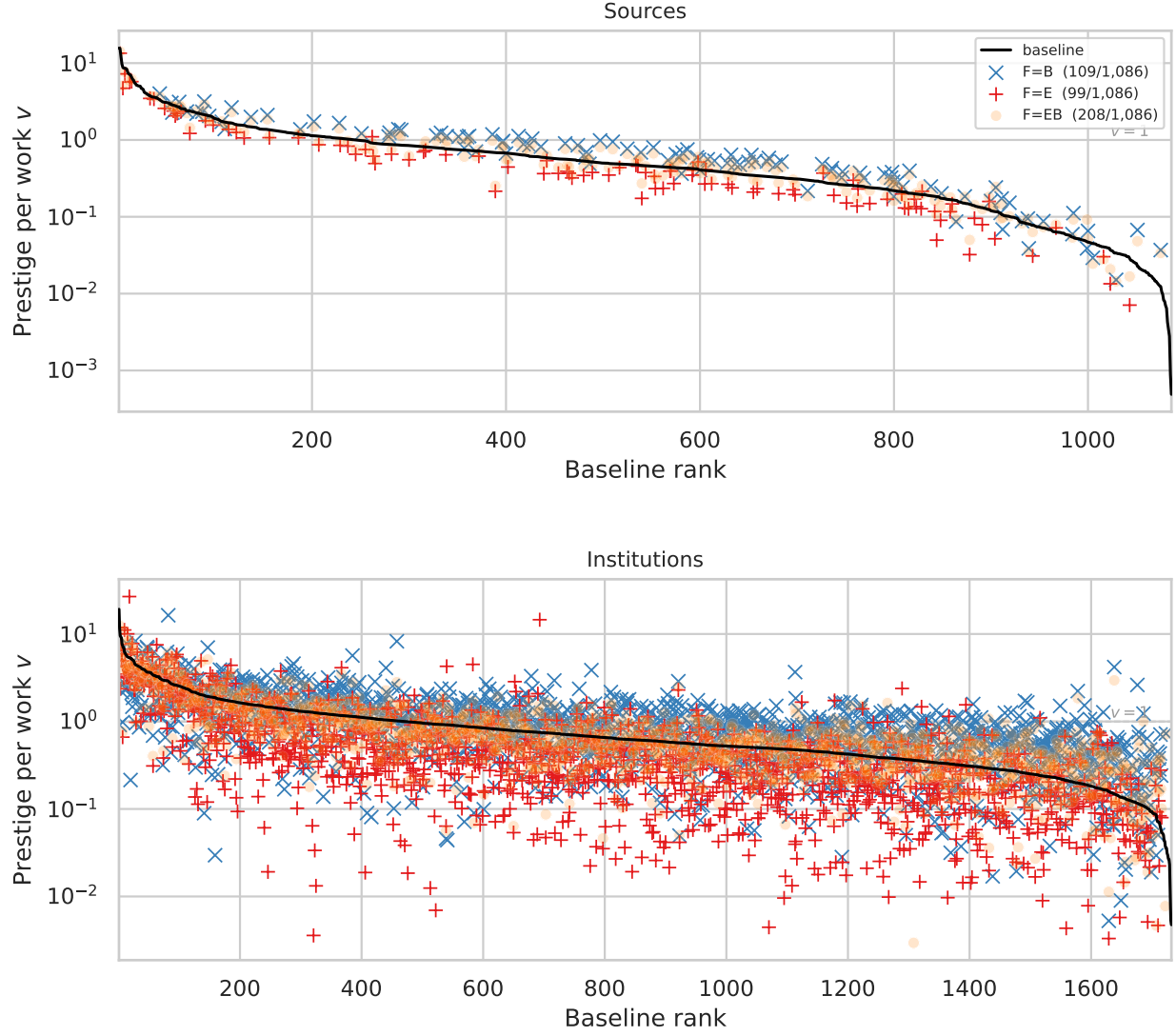


Figure 2: Prestige per work  $v$  against baseline rank. Top: sources. Bottom: institutions. Black line: baseline ( $F = A$ ). Red +:  $F = E$  ( $N_s = [FILL]$ ). Blue x:  $F = B$  ( $N_s = [FILL]$ ). Dashed line:  $v = 1$  (corpus average). Parameters:  $t_x = 5$ ,  $\tau_U = \tau_S = 20$ ,  $\rho = 0$ ,  $\mathbf{m} = 0110$ ,  $\alpha = 1$ .

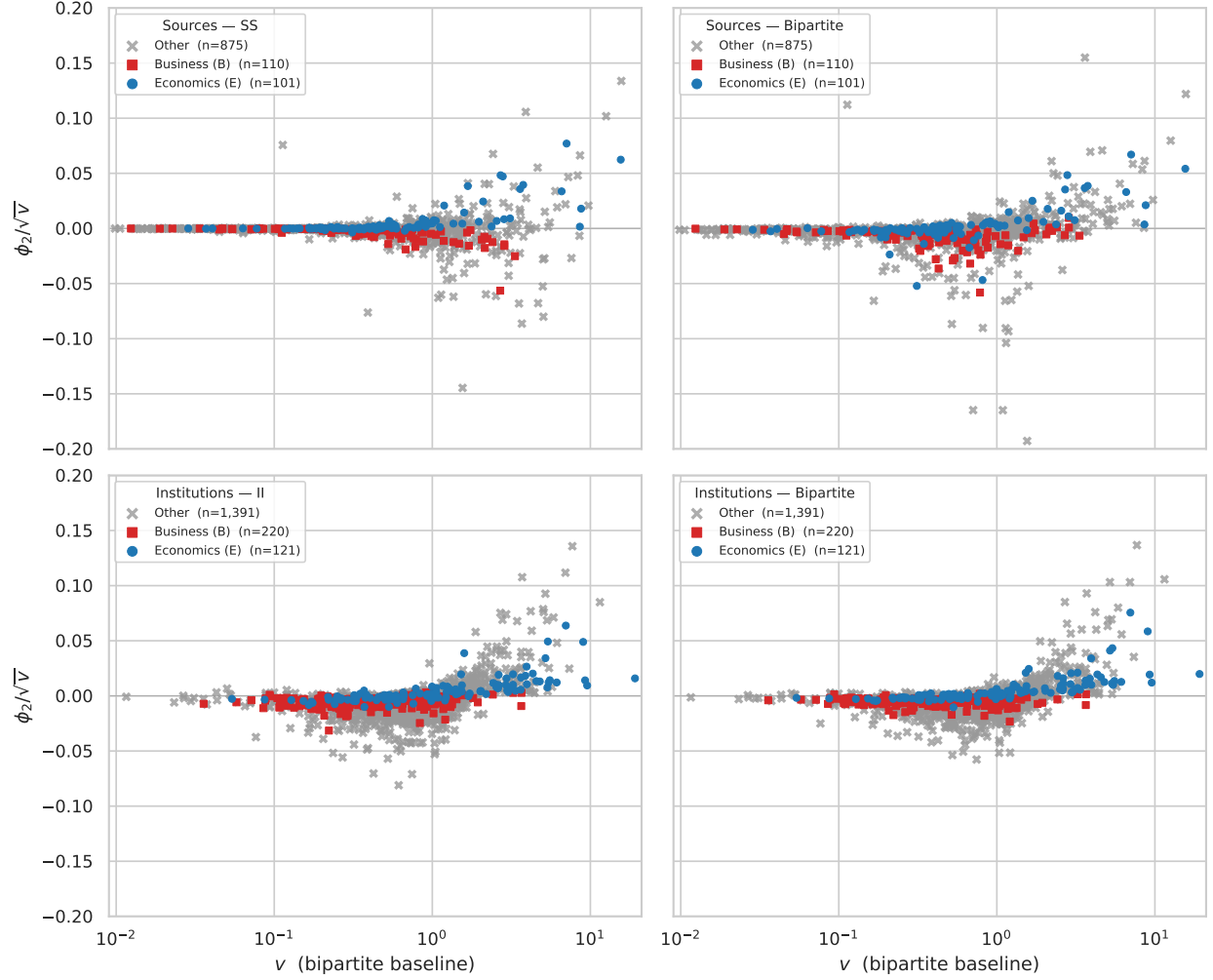


Figure 3: Community measure  $\phi_{2,p}/\sqrt{v_p}$  against prestige per work  $v_p$ . Top row: sources. Bottom row: institutions. Left column: SS mode  $\mathbf{m} = 1000$ . Right column: bipartite  $\mathbf{m} = 0110$ . Sources coloured by field: blue circles = Economics ( $E$ ); red squares = Business ( $B$ ); grey crosses = other. Institutions coloured by field-network membership (E-only, B-only, other).  $x$ -axis: bipartite baseline  $v$ . Parameters:  $t_x = 5$ ,  $F = A$ ,  $\tau_U = \tau_S = 20$ ,  $\rho = 0$ ,  $\alpha = 1$ .

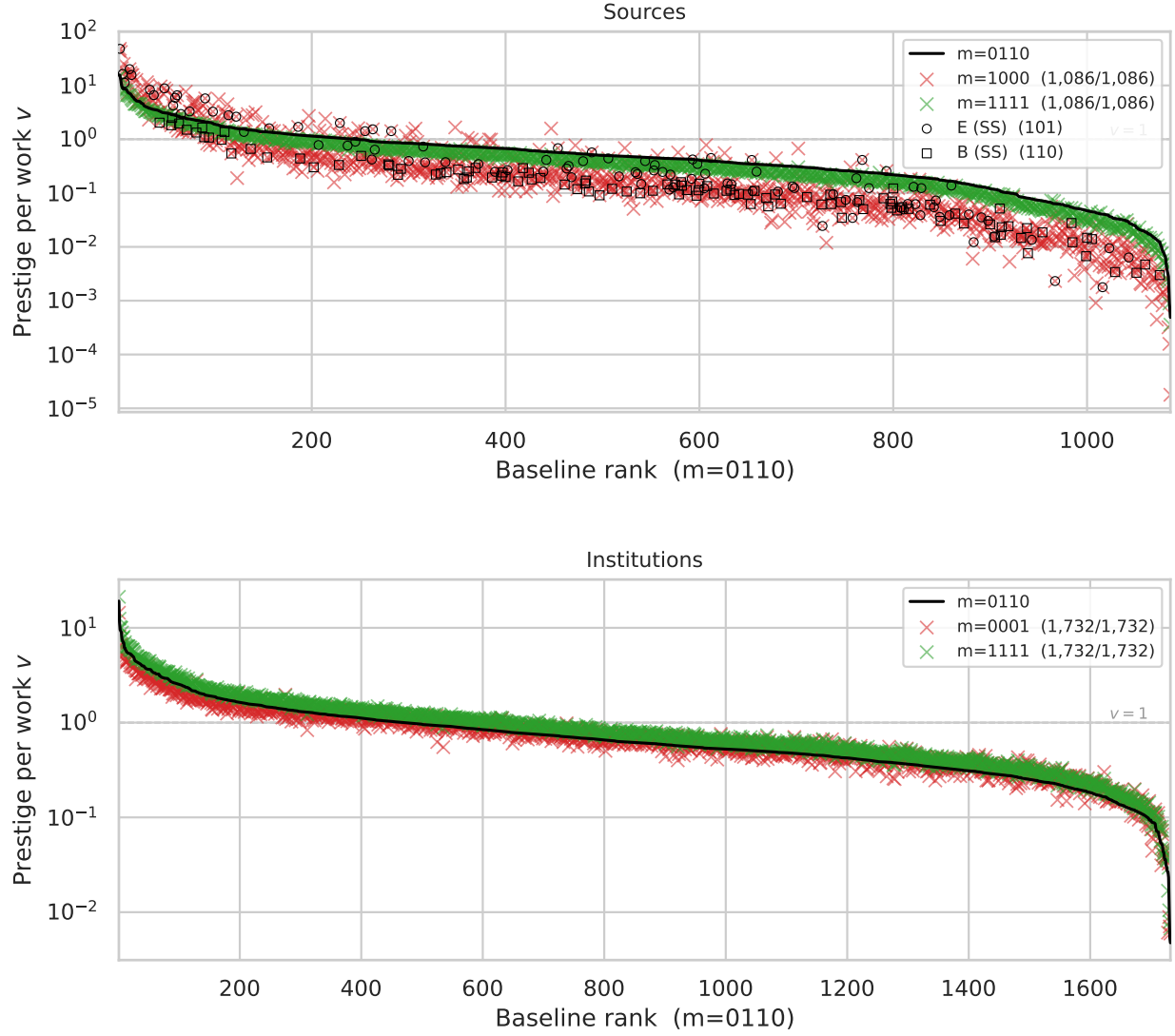


Figure 4: Prestige per work  $v$  against bipartite baseline rank. Top: sources. Bottom: institutions. Black line: bipartite baseline  $\mathbf{m} = 0110$ . Red  $\times$ :  $\mathbf{m} = 1000$  (top) /  $\mathbf{m} = 0001$  (bottom). Green  $\times$ :  $\mathbf{m} = 1111$ ,  $\chi^* = [FILL]$ . Open circles ( $\circ$ ) and squares ( $\square$ ) overlaid on the 1000 series (top) and 0001 series (bottom) mark Economics and Business units respectively, classified by field-network membership. Parameters:  $t_x = 5$ ,  $F = A$ ,  $\tau_U = \tau_S = 20$ ,  $\rho = 0$ ,  $\alpha = 1$ .

## 5 Discussion

**Face validity.** [Placeholder.]

**Content validity.** [Placeholder.]

**Criterion validity — concurrent and predictive.** [Placeholder.]

**Construct validity.** [Placeholder.]

## 6 Prospects

[Placeholder.]

## 7 Addenda

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